Appendix A — Mathematical Structure

A.1 — Collapse Attractor Set The collapse attractor set is defined by:

$$\mathbb{P} = \{2, 3, 5, 7, 11, \dots\}$$

A.2 — Collapse Attenuation Kernel The attenuation function is:

$$A_p(x) = \frac{1}{(|x-p|^\theta + \sigma^2)^\kappa} + \rho \cdot \log(|x-p|+1)$$

A.3 — Collapse Amplitude Each prime-indexed attractor has:

$$f_p(x,t) = \alpha_p \cdot \cos(\omega_p t + \delta_p)$$

A.4 — Information Field Gradient Complexity field:

$$I(x,t) = \log \mathcal{C}[\psi(x,t)], \quad \nabla_\theta I(x,t) = \left(\frac{\partial I}{\partial x}\right)^\theta$$

Structural Commentary (SOTP Disclosure)

Note on Structural Modulation: Auxiliary terms like $R(x,p)$ serve not as heuristic corrections but as structurally resonant components. Their role aligns with principles found in effective field expansions, contributing to stability and scale coherence while remaining agnostic to deeper generative formulations.